

Problems with classical computers and important deviations

Problems with classical computer are two-folds:

1. How to use its power for general-purpose small computing jobs, thereby having a cost-effective solution and raising the system productivity?

2. How to raise its power, performance and capacity to tackle extensive, complex numerical jobs (for example, design of supersonic aircraft and modeling of global weather) where if a serial computer is used, it may take many years to solve the problem?

The solution to the first problem came in the year 1960, with the introduction of time-sharing multi-user concept. This was based on the philosophy of utilisation of slowness of humans compared to computers, so that if one user is thinking, the computer can be used by other users (resource sharing by time slice). This provided a means of distribution of the cost of computing over many users.

Another early solution to the first problem was 'batch system,' which remained dominant where a large amount of data was processed with minimal human interaction (one operator). But as it was not interactive type, it lost itself to the time-sharing system.

One of the answers to the second problem gave birth to parallel computing, which is the ultimate aim of the fifth-generation computing system. Some parallel computers are already in operation in the world. Parallel computing speeds up the operation. With this thing in mind, the concept of optical computer was developed.

In optical computers, it is the light that will carry the signals; and in universe, it is the light that has the ultimate speed. Accordingly, non-linear optics emerged as the new frontier of science and technology.

Another important deviation from classical computer that emerged due to technological growth and demand, was the design of brainy computers. The chemical computer is a bold step in formulating the brainy computer. Optical and chemical computers are now merged under a new field of electronics known as molecular electronics.

laws. Pat Gelsinger's vision of expanding Moore's laws includes Intel's 90nm (and now 45nm) fabrication process. Although several alternative technologies, namely, quantum computing, bio computing, molecular electronics and chemical computing, are under investigation as possible replacement of digital computing, the year 2010 may achieve the landmark of a billion-transistor IC—another leap forward in IC technology.

However, high-density, high-speed chip design is a major challenge. It is predicted that by 2016, high-performance ICs will contain more than 8.8 billion transistors on an area of 280 mm², which is more than 25 times that of today's chips. Power dissipation will then double, requiring elaborate arrangements of heat-sinks. As the transistor dimensions decrease, so does the channel length. With decrease in channel length, the speed will increase. Many research findings reveal that heat dissipation will ultimately limit the chip integration/density and operating frequency. So going with Moore's law beyond 2016 does not

look possible.

The new technologies that are under active consideration under such a scenario are molecular electronics and quantum computing. Both the technologies are still in conceptual stages, although a few limited implementations are reported.

Molecular electronics

The basis of information processing by computers lies in the representation of two binary states or bits, namely 0 and 1. In conventional computing, binary states are represented by 'on' and 'off' states of transistors. In molecular electronics, the characteristics of a molecule like molecular state and spin orientation are proposed to represent the bits. For example, the positive spin and the negative spin of a molecule may represent bit 1 and 0, respectively, or vice versa. A bit of information can then be stored in the space of an atom or molecule.

The subject of nano computing started with the concept of molecular electronics followed by chemical computing. Molecular electronics emerged

as an important area of research and application during 1980s. The definition of molecular electronics is not unique and simple. Scientists differ even within a country.

A leading scientist of the field divides molecular electronics into two main themes: molecular materials for electronics (MME) and molecular-scale electronics (MSE). The topic of molecular materials for electronics deals with the use of macroscopic properties of organic materials in devices. In the near term, conductive polymers will lead to development of novel electronic devices and organic materials with pronounced non-linear optical properties will find application in electronics. A simplistic extrapolation of the reduction of time eventually leads to the molecular-scale electronics.

Prof. Bloor further observed that "many regard the quest for molecular-scale devices as true molecular electronics. However, the distinction between MME and MSE is somewhat arbitrary and both need to be considered as constituent parts of molecular electronics if the topic is to grow and prosper."

Ashwell, Sage and Trundle broadened the definition "from electronics at molecular level to include molecular materials with potential electronics and photonic applications."

Peterson defined that "in the most general sense, molecular electronics covers the use of molecular (and hence essentially organic) materials to perform signal processing or transformation function."

The famous Link programme of Britain defined molecular electronics as "systematic exploitation of molecular, including macro molecular, materials in electronics and related areas such as photo electronics."

Molecular electronics therefore explores the potential application of organic materials and non-linear optics in the field of electronics. It is a highly interdisciplinary field and its prospects depend on the successful interaction and cooperation between scientists of